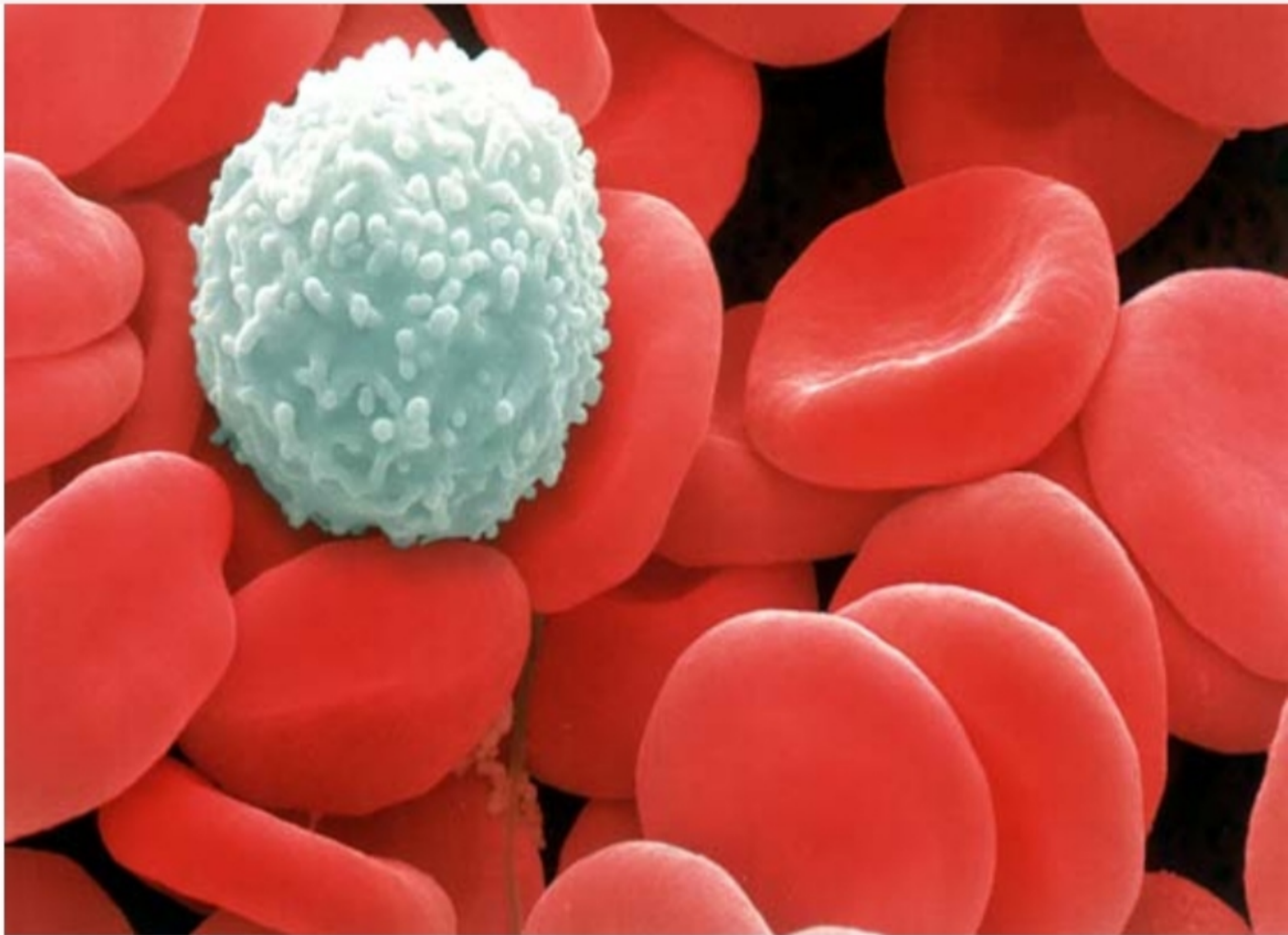


blood

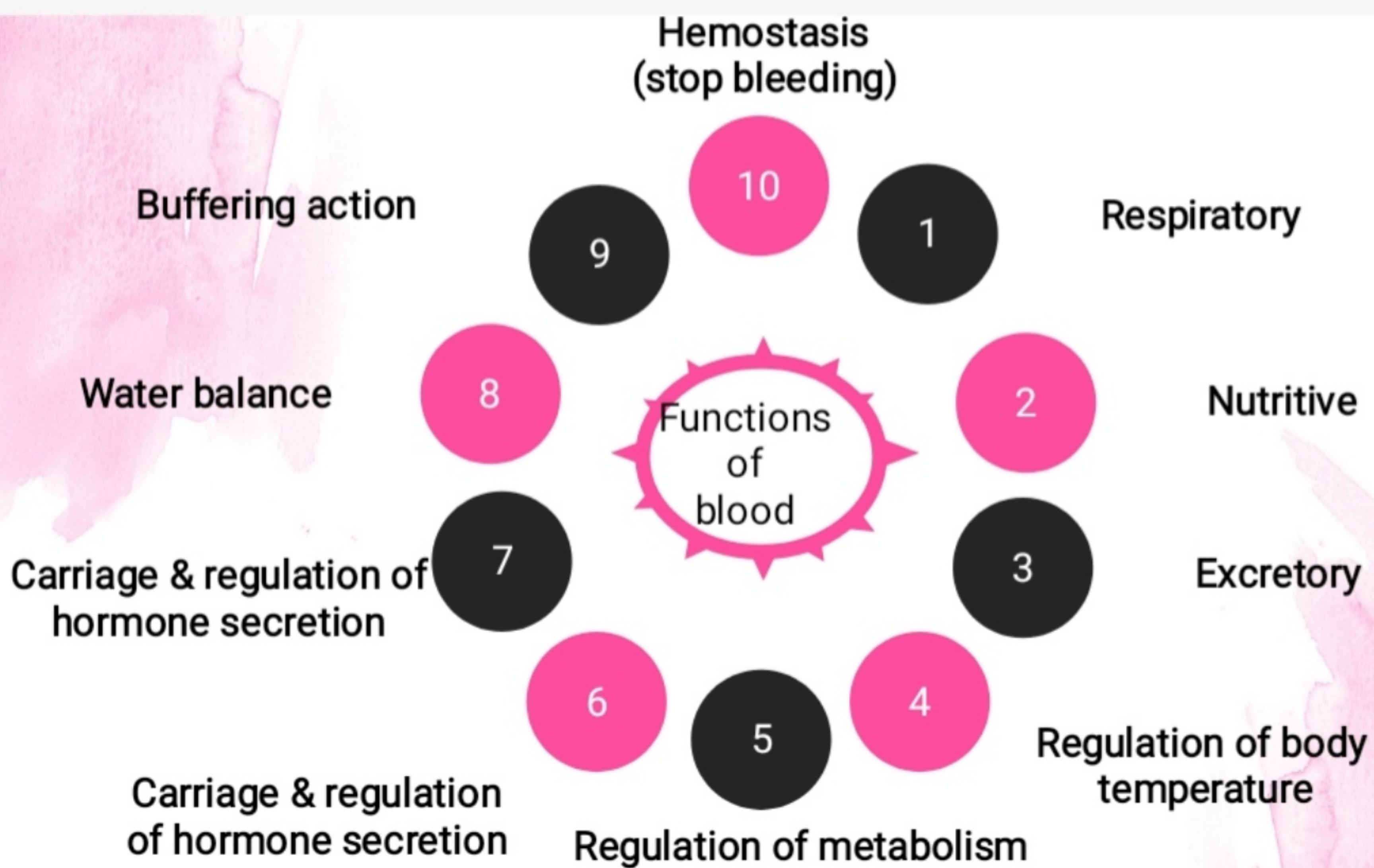


Blood (Hematology)

Blood, lymph & tissue fluids form the internal environment of the organism, washing all its cells and tissues. The internal environment is marked by a relative constancy in its composition and physico-chemical properties (**homeostasis**). Gross variation in the internal environment is fatal.

Blood

Total amounts of the blood about **6-8 %** of the body weight (3.3 L/m² body surface area), less in female than male by **7%**. Blood is distributed in the body as the following: **60%** in systemic veins, **20%** in the blood vessels of the lung & **20%** in heart, arteries & capillaries.



Functions of blood

Blood keeps a constant environment around the body cells via its different functions, which are summarized briefly in the following:-

- 1) **Respiratory:** blood carries oxygen from the lung to the tissues and carbon dioxide from the tissues to the **lung**.
- 2) **Nutritive:** blood carries foodstuffs e.g. glucose, amino acids, fats, vitamins and minerals from the **digestive tract** to the tissues.
- 3) **Excretory:** blood remove waste products e.g. urea, uric acid and excess foodstuffs from the tissues to the excretory organs mainly the **kidneys**.

Functions of blood

4) **Regulation of body temperature:** blood regulate body temperature through:-

a) Blood distributes temperature equally all over the body. It can rapidly transport heat from deep tissues to the surface of the body or the opposite.

b) Blood inform the **hypothalamus** (which has sensitive cells to very little changes in blood temperature, $\pm 0.01^{\circ}\text{C}$). The hypothalamus responds in a way that keeps normal body temperature (through its control of the autonomic nervous system). So, in cold weather the skin vessels are constricted and the heat loss to atmosphere is decreased. In hot weather the reverse occurs to help heat loss into the atmosphere.

Functions of blood

5) Regulation of metabolism: blood carries hormones, enzymes, vitamins and other substances from the site of production or storage to the tissues that need it in the metabolism.

6) Defense: blood defends the body by two mechanisms:-

a) Antibody formation: blood contains defensive substances (e.g. antibodies and antitoxins), which protect the body against harmful bacteria and poisons.

b) Phagocytosis: the white blood cells play an important role in the defense against bacteria through engulfing and digesting them.

Functions of blood

7) Carriage & regulation of hormone secretion:-

- a) Hormones are carried only by blood from ductless glands to the tissues which act on them.
- b) Hormonal secretion is regulated by blood by a method of feed- back mechanism.

8) Water balance: blood removes excess water from tissues to kidneys and sweat glands. Also, it can give water to tissues if they are dehydrated.

Functions of blood

9) Buffering action: the blood containing many important buffer systems, which help to keep the **pH** of blood and tissues almost constant. Each system is composed of a **weak acid and its salt** (e.g. H_2CO_3 & NaHCO_3). The acid neutralized any changes of blood pH towards alkalinity, while the base neutralized changes towards acidity.

10) Hemostasis (stop bleeding): blood can prevent bleeding from injured blood vessels by many mechanisms (e.g. **coagulation of the blood**).

Physical properties of the blood

1- Color: arterial blood is **red** due to oxy-hemoglobin, while venous blood is **bluish** due to reduced hemoglobin (Hb). Under **abnormal conditions**, the blood may have other colors e.g. cherry red due to the presence of carboxy-hemoglobin or brownish due to the formation of met- Hb.

2- Specific gravity: specific gravity is 1.000 for water, 1.030 for plasma, 1.060 for whole blood and 1.090 for blood cells. Specific gravity of the blood cells is the bases of two blood tests (**ESR and PCV**).

Physical properties of the blood

3- Viscosity: viscosity is 1 for water, 2 for plasma & 5 for whole blood. Blood viscosity is due to **RBCs and plasma proteins**. Blood viscosity is necessary for maintaining **normal arterial blood pressure** and preventing rapid flow of blood, which give a chance for the exchange between blood and tissues.

Physical properties of the blood

4- Osmotic pressure (O.P.): it is due to **crystalloid O.P.** (mainly due to NaCl = 5000 mmHg) and **Colloid O.P.** (due to plasma proteins = 28 mmHg). O.P. of the blood controls the exchange between blood and tissues (through the process of filtration and absorption).

5- Reaction (pH): the pH is 7.37 of Venus blood & 7.4 for arterial blood. Blood pH is kept constant by many chemical and physiological mechanisms. Gross changes in blood pH are fatal because range of pH which are suitable for life is narrow (7-7.8).

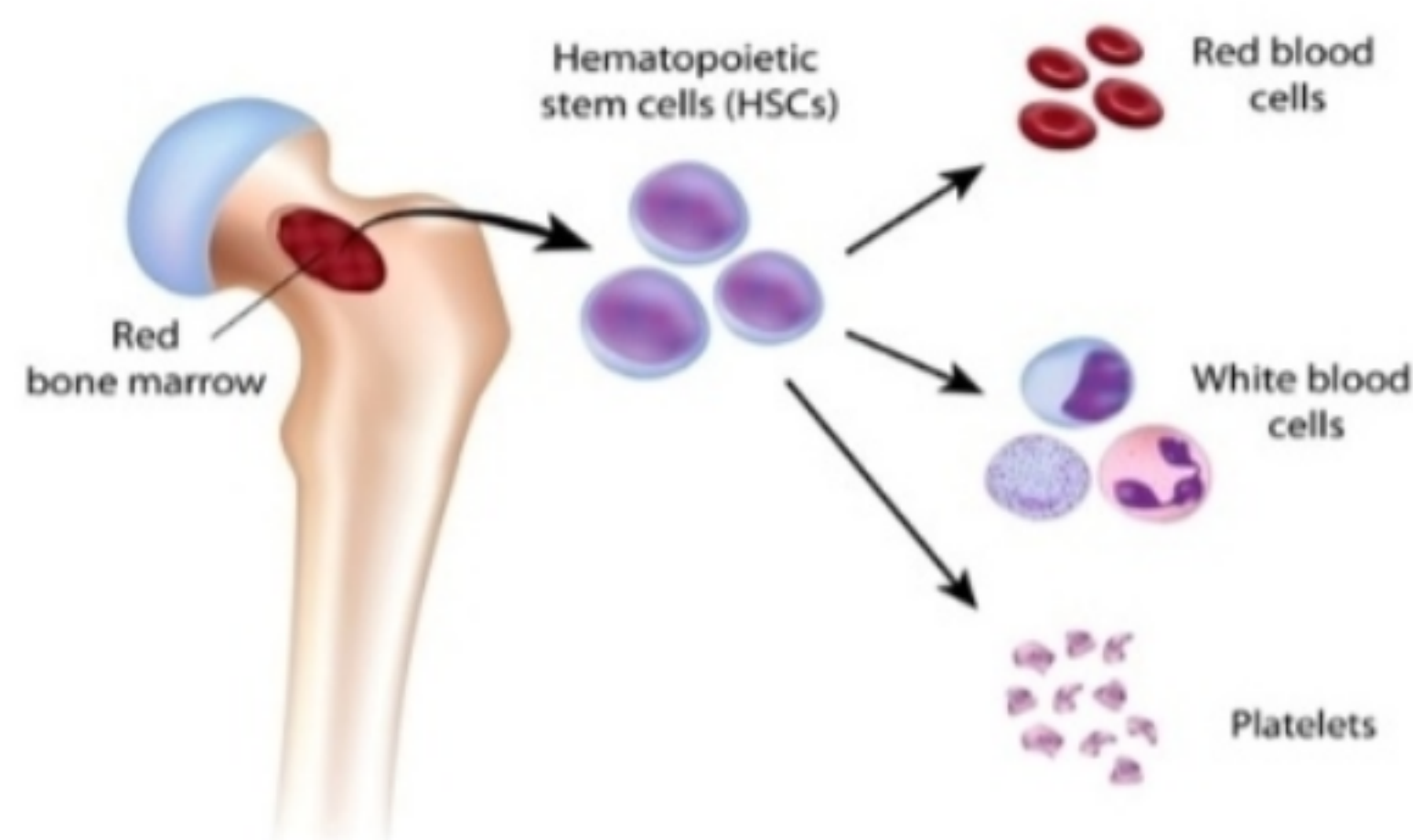
Bone marrow

In the **adult**, the red blood cells (RBCs), most of the white blood cells (WBCs) and blood platelets are formed in the bone marrow. The bone marrow is one the **most active organs** in the body and one of the largest organs (approaching the size and weight of the liver).

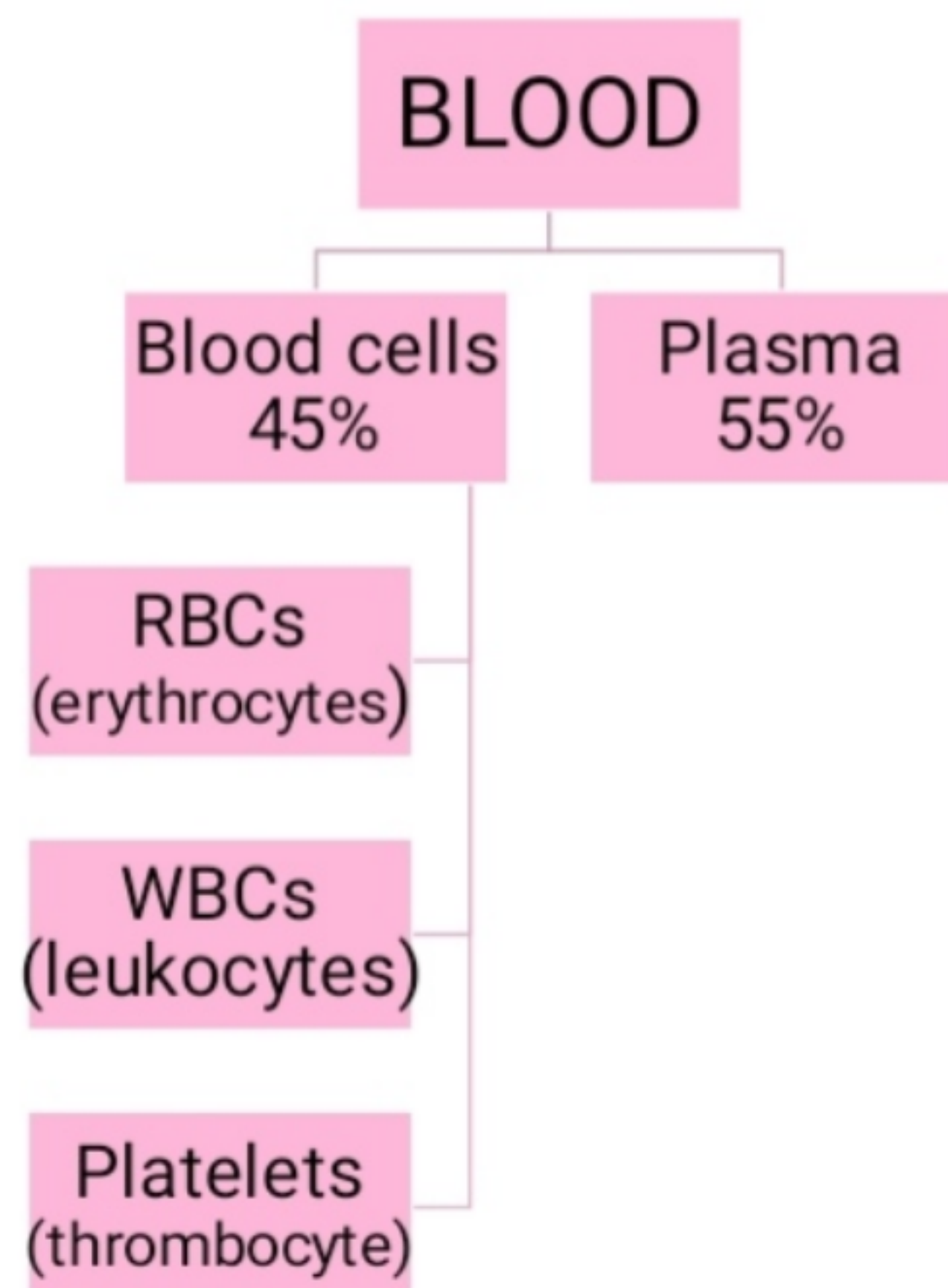
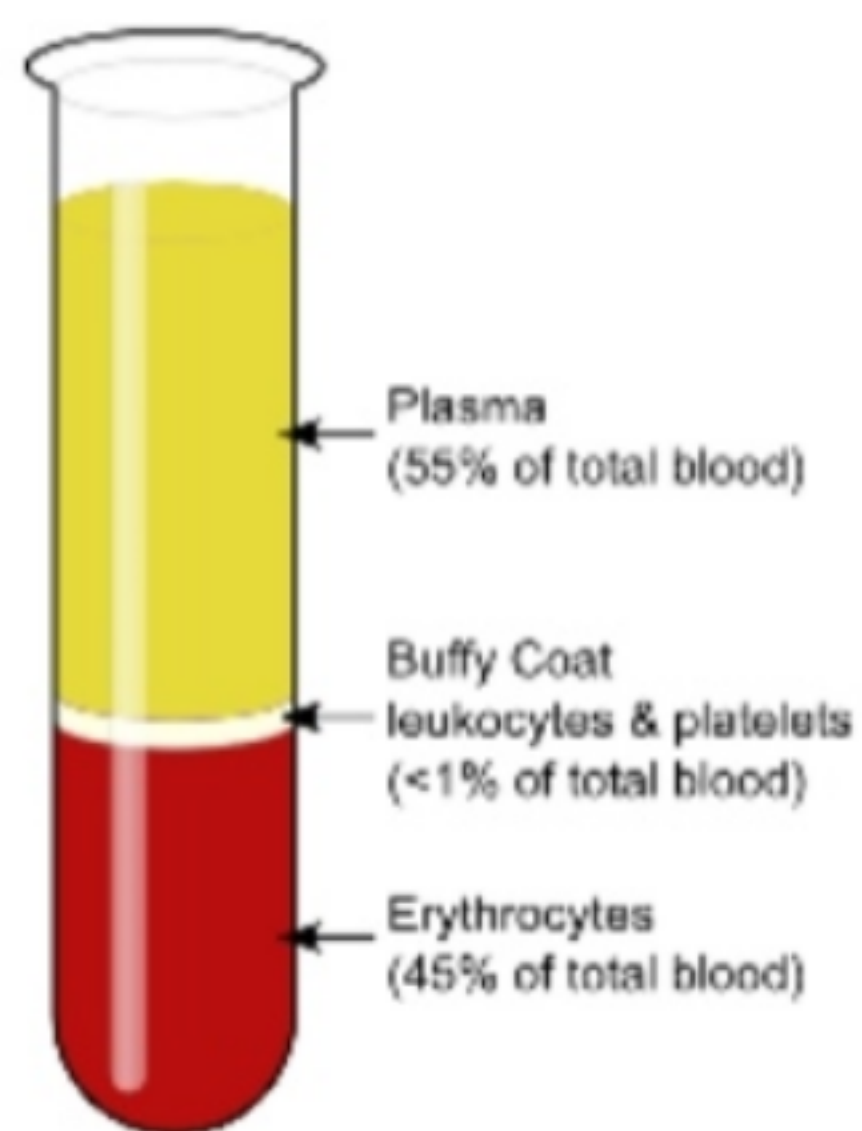
Active bone marrow is **red in color** and with increasing in age fat is deposited in the bone marrow of the long bone and become inactive (not producing blood cells) **yellow bone marrow**. There is a stem cell for each type of blood cell (except neutrophil & monocyte from one mother cell).

Bone marrow

Normally, **75%** of the bone marrow cells belong to **WBCs**-producing & only **25%** to **RBCs** and blood platelets due to short life span of the WBCs.



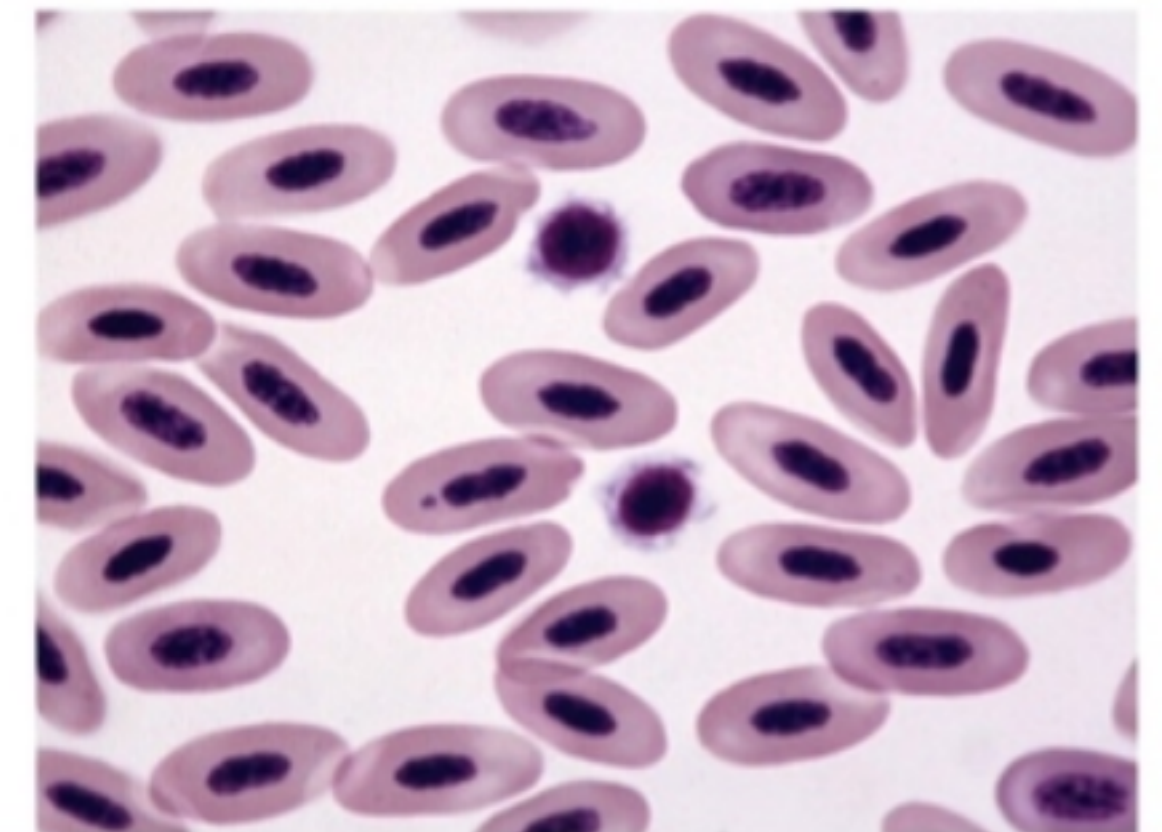
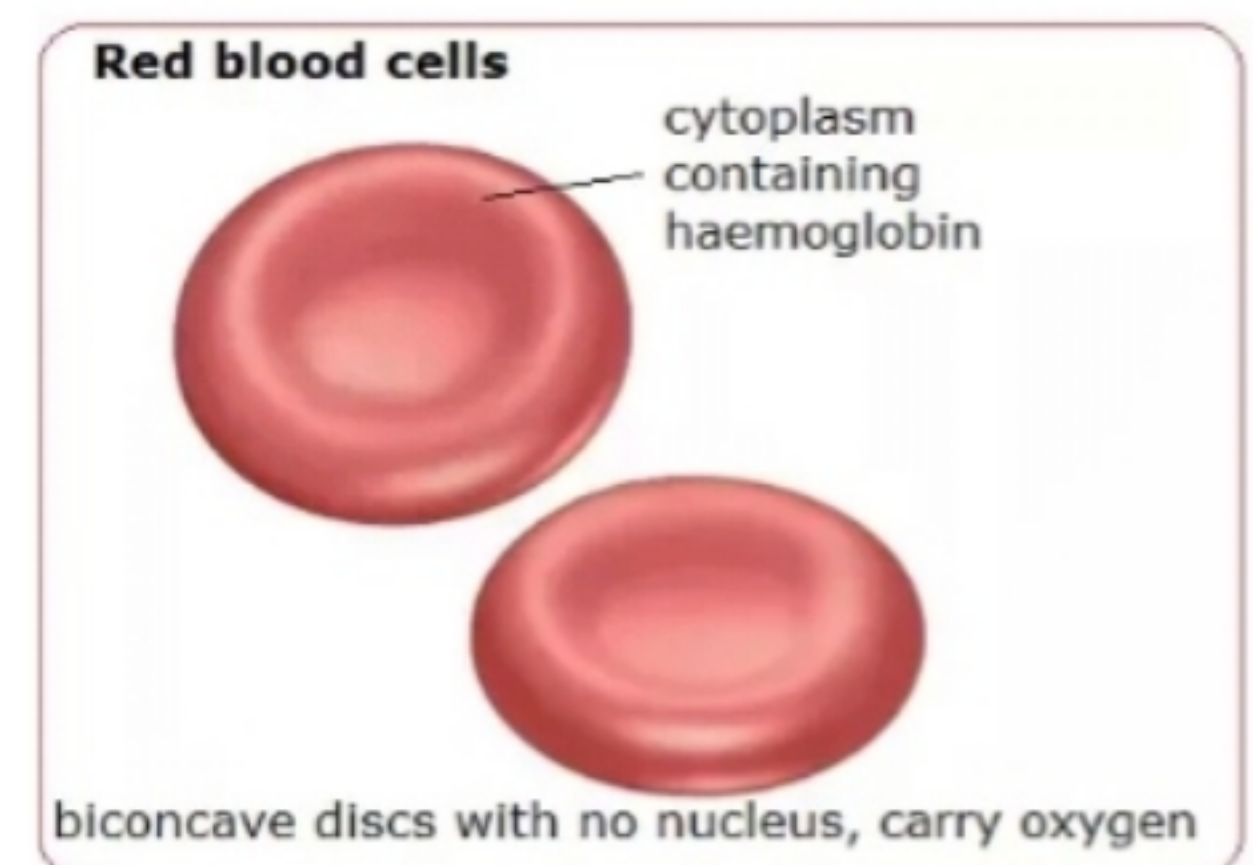
Composition of blood



Red Blood Cells (RBCs or Erythrocytes)

Erythrocyte of mammals is **circular** and **non-nucleated** (except in birds, reptiles & fishes are ovoid and contain a nucleus).

They appear as a **biconcave**. They are non-motile, **Red in color** (due to Hb). Its diameter **4-8 μm** . The total surface area of the RBCs in the body is about 62m²/kg B.W.



Red Blood Cells

Site of formation:

It is formed early from the **yolk sac** then from the **fetal liver, spleen & lymph nodes** and when the bones are formed the RBCs are produced by **bone marrow** till death. In birds, the spleen also forms small amount of the RBCs.

Under certain **pathological** conditions (as destroy or fibrosis of the bone marrow) in post-natal life, liver, spleen and lymphnodes may again assume their fetal function of producing RBCs.

Functions of erythrocyte

A

RBC membrane

B

RBCs content

Functions of erythrocyte

1) RBC membrane:-

- a) **Plasticity** of this membrane make the erythrocyte resist rupture (hemolysis), when pass through the narrow blood vessels (the membrane plastic not elastic).
- b) The **biconcavity** of the membrane gives the RBC a large surface area, which helps the **gas exchange**.
- c) It keeps the **carbonic anhydrase enzyme inside** the RBC, otherwise it would be lost in the urine. This enzyme is important for **Co₂ carriage** in the blood.

Functions of erythrocyte

d) It is **semi-permeable** and so it regulates the distribution of anions (as Hb^- , Cl^- ; HCO_3^-) and cations (as Na^+ ; K^+) inside and outside the RBC.

It is not permeable to K^+ , Hb^- , Na^+ (K^+ is kept inside and Na^+ outside the RBC). It is permeable to H^+ and Cl^- . This is important for keeping the **pH of the blood constant**, when carbon dioxide is added by the tissues.

Functions of erythrocyte

e) It keeps the **Hb inside the RBC**, this is very important because if Hb is free in plasma losses its functions and causes the following problems:-

- 1- May be precipitated in the kidneys as acid hematin which blocking them and inducing anuria.
- 2- Would increase the osmotic pressure of plasma proteins 5 times, which would prevent tissue fluid formation.
- 3- It was liable to be engulfed and destroyed by the macrophage.
- 4- May be filtered from the kidneys and lost in the urine.

Functions of erythrocyte

2) RBCs content:-

a) Hemoglobin (Hb): It have the following functions:-

- 1- Oxygen carriage from the lungs to the tissues and the opposite for the Co_2 .
- 2- Considered as a buffer system in the blood.

b) Carbonic anhydrase enzyme, which is important in the Co_2 carriage.

c) Histaminase enzyme, which inactivate histamine after performing its function.

Composition of erythrocyte

1) Water: 65%.

2) Solids: 35%

a) Hb: 34% (about 95% of all solids)

b) Others: 1% include the following materials:-

1- Proteins: found in the stroma and cell membrane.

2- Glucose, for energy.

3- Lipids: phospholipids, cholesterol and neutral fat.

4- Vitamins, used as coenzymes.

5- Enzymes: as cholinesterase, phosphatases, carbonic anhydrase, peptidases & glycolysis enzymes.

6- Minerals: such as P, Cl, Mg, Na; S. Sodium is the principle cation in extracellular fluid, while potassium is the principal cation of erythrocyte.

Erythropoiesis

Sites of RBCs formation:-

In early fetal life, nucleated RBCs are produced in the **yolk sac**. Later on, **liver, spleen and lymphnodes** are involved. In the last stages of the fetal life as well as after birth, the **bone marrow** takes the function of the RBCs formation.

In birds, **spleen** also forms RBCs to a small extend during life.

Under certain pathological conditions in postnatal life (as in destroy or fibrosis of the bone marrow), liver, spleen and lymphnodes may again assume their fetal function of erythropoiesis.

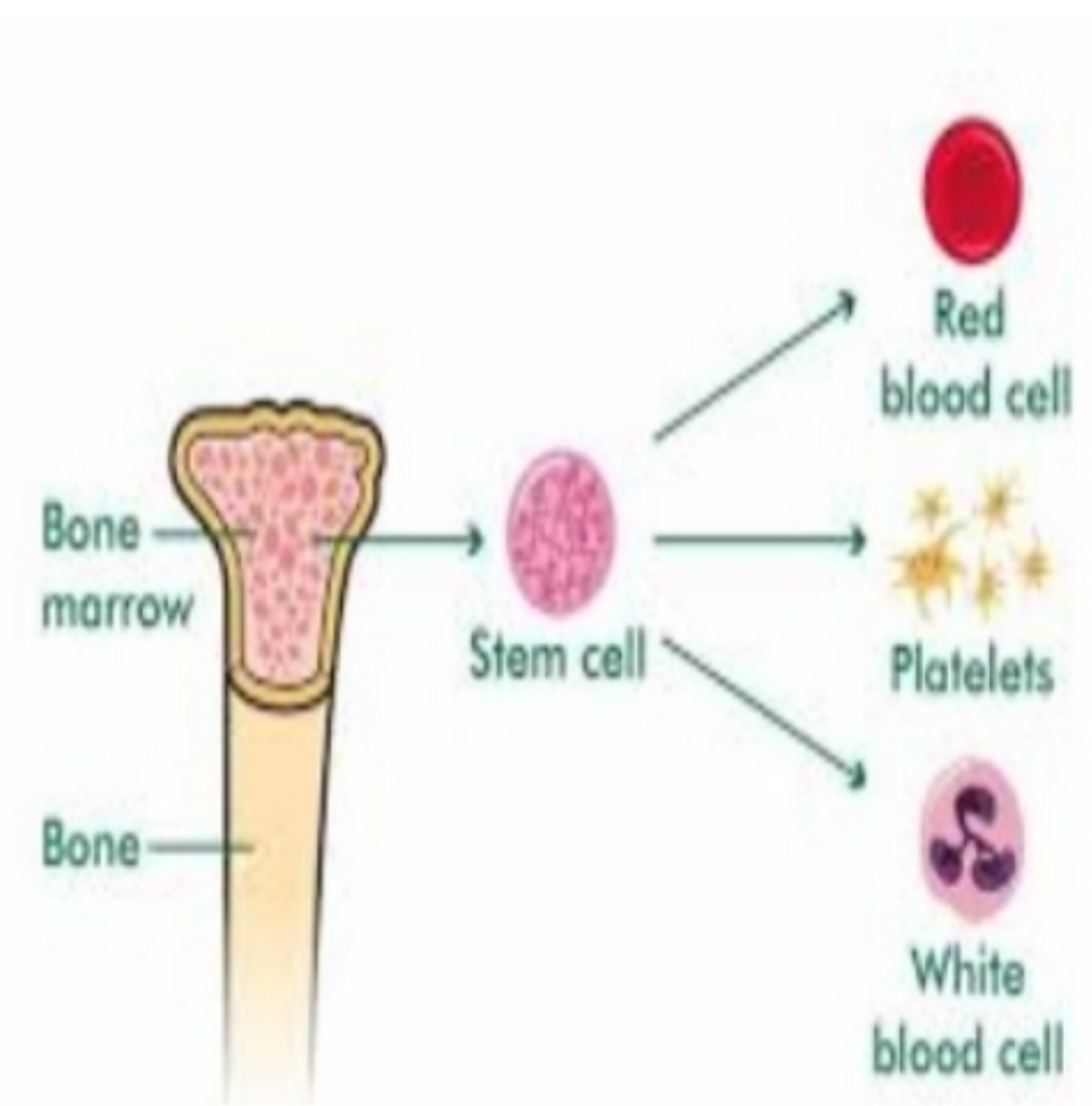
Sites of RBCs formation:-

In young age, all the bone marrow is red, but in adult the shafts of long bones become filled with yellow fatty marrow (does not produce RBCs). The upper humerus and femur are still red and produced RBCs. In adults RBCs are formed in the flat bones (e.g. skull, ribs, vertebrae, sternum, scapulae and clavicles).

Developmental stage

Stem cell → Hemocytoblast → Erythrocytoblast
(beginning of Hb appearance in the cytoplasm) →
Normoblast (34% Hb) → Reticulocyte (the
nucleus fragmented) → Erythrocyte (enter the
blood circulation without nucleus except in birds,
reptile and fishes).

In some **pathological condition** (as
hemorrhagic anemia), when the body needs
extra formation of RBCs, the **reticulocyte** appear
in the blood stream in a noticed percentage
(over the normal percentage, which are **1-3 %**).



Factors affecting erythropoiesis

RBCs formation is a continuous process. Certain factors are essential for perfect RBCs regeneration (the deficiency of any of them leads to a type of anemia):-

1) Oxygen tension (pressure):-

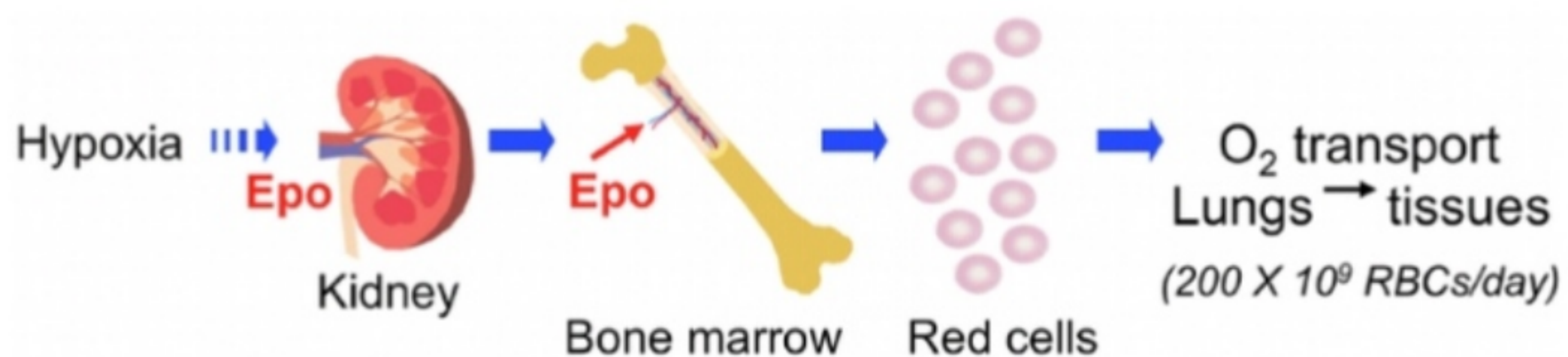
Lowering of the oxygen tension in arterial blood (as during living at **high altitude**), stimulate the release of **erythropoietin hormone** which stimulate the bone marrow to produce **more RBCs**. While, increasing the blood oxygen tension (as in breathing pure oxygen), inhibits RBCs formation by the bone marrow.

Factors affecting erythropoiesis

2) Kidney:-

The **juxtaglomerular cells (J.G.C)** of the kidney responded to **hypoxia** by producing **renal erythropoietic factor**, which combined with **α - globulin** (produced by the liver under the effect of hypoxia) to form the **erythropoietin hormone** which stimulate the **bone marrow** to produce more RBCs.

The kidney produces about 85% of the erythropoietic factor while the liver and the any hypoxic tissues produced about 15%.



Factors affecting erythropoiesis

3) Food factors:-

a) **Proteins:** proteins specially the **essential amino acids** are more effective in erythropoiesis.

b) Trace elements:-

1- **Iron:** It forms about **0.334%** of the Hb molecule.

Factors affecting erythropoiesis

Absorption and transportation of the iron:-

Food iron is ingested as $\text{Fe}(\text{OH})_3 \rightarrow$ free **Fe^{+3}** (by gastric HCl) $\rightarrow$ reduced to **Fe^{+2}** (by **Vit. C & Ca^{+2}**), which absorbed by the intestinal cells, then oxidized to $\rightarrow \text{Fe}^{+3}$ Apoferritin $\rightarrow$ **Ferritin** (stored form in the **intestinal cells, liver, spleen and bone marrow**). It can be reduced again to ferrous form and transported from the intestinal cell to the blood by a β - globulin (produced by the liver, called **transferrin**), to the **bone marrow**.